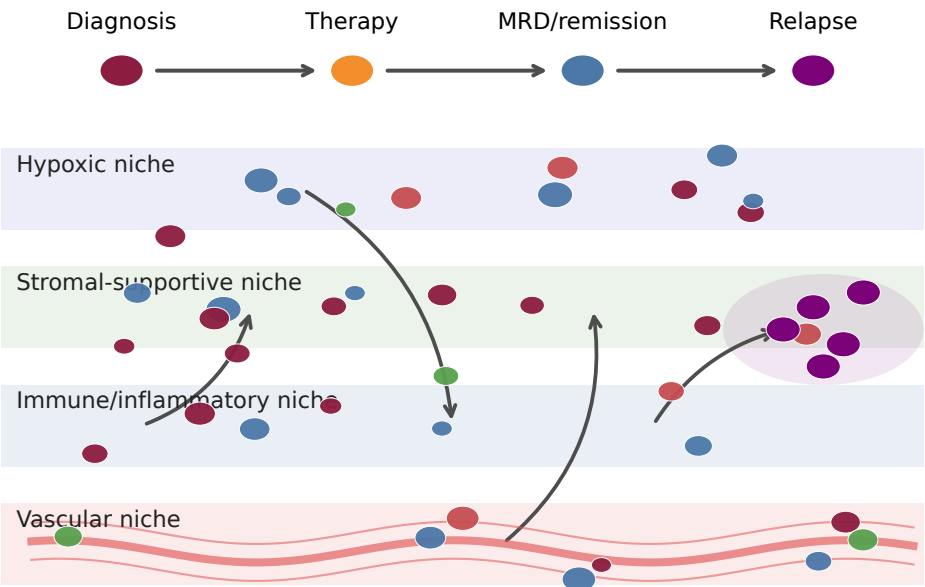


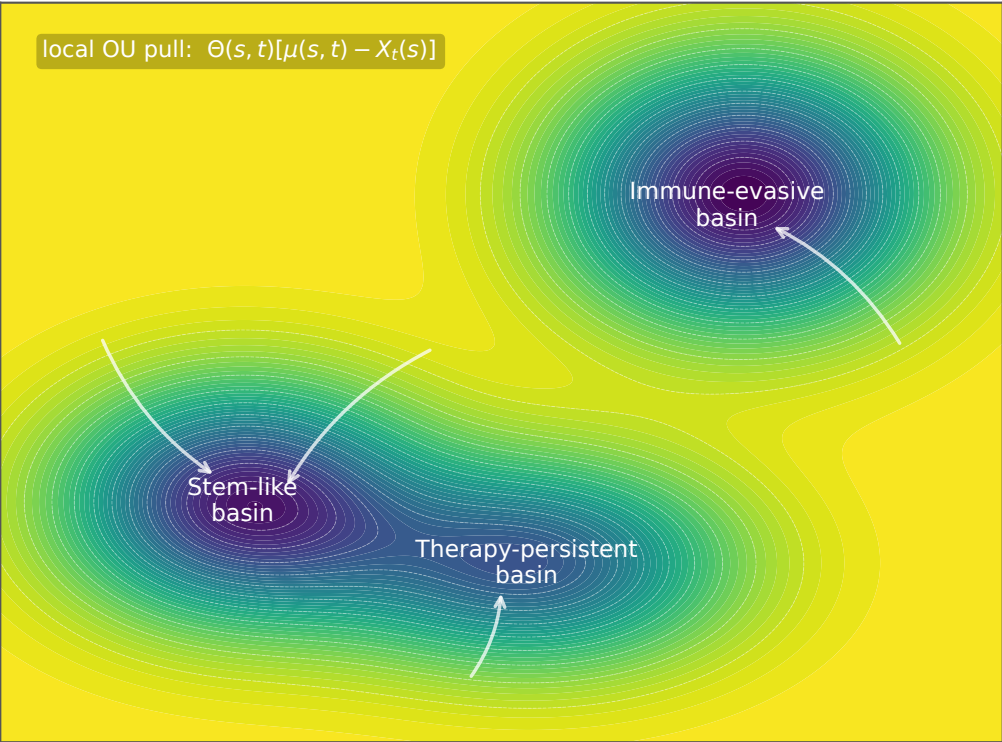
Spatial therapy-aware OU-Lévy-branching ecology of pediatric leukemia evolution

A Dynamic spatial leukemia evolution

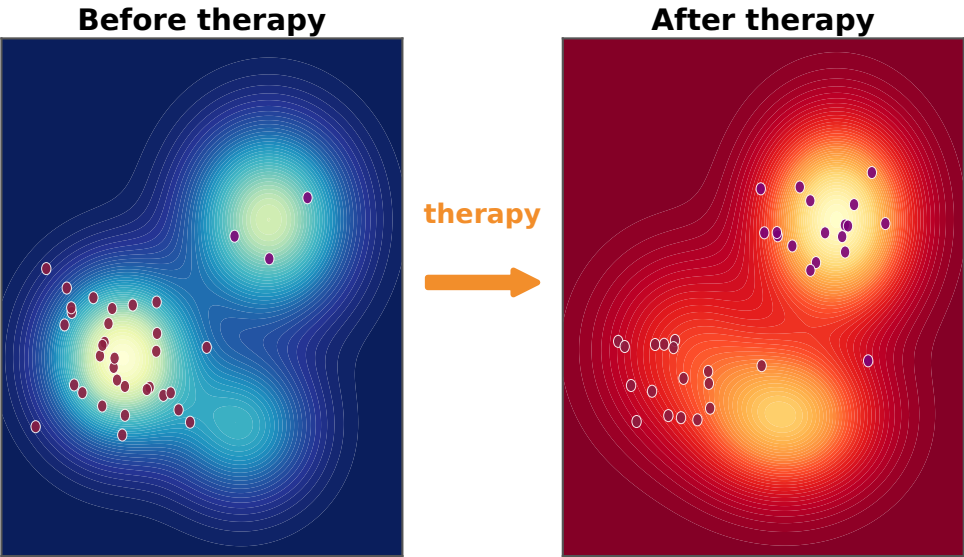


Therapy-driven spatial reconfiguration

B Spatial ecological attractor basins



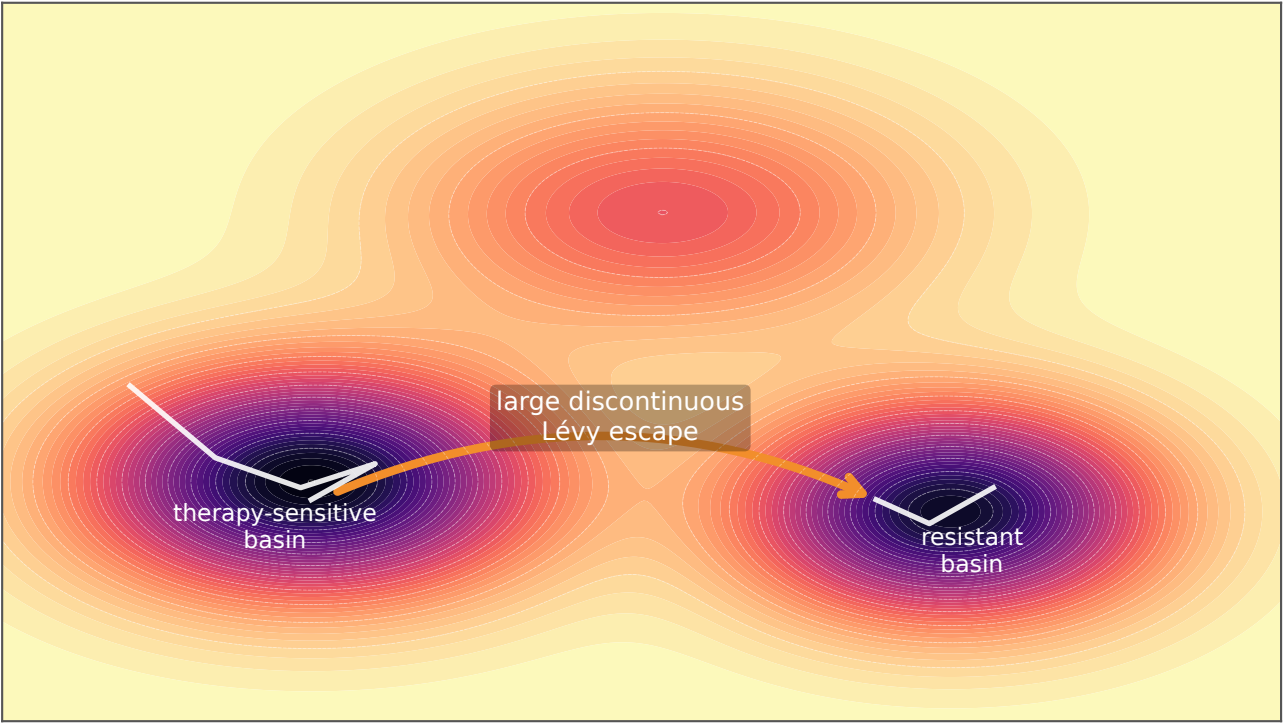
C Therapy-induced ecological perturbation



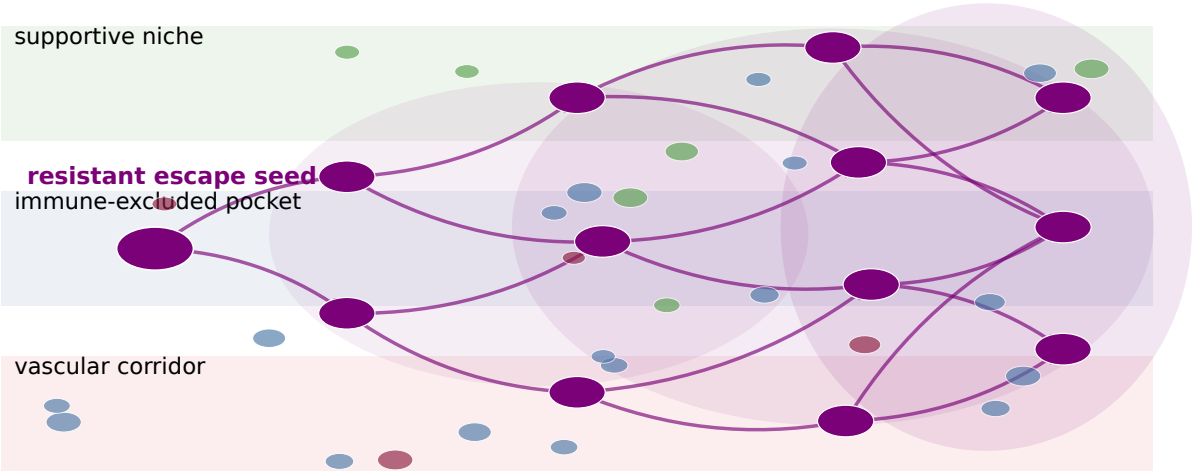
attractor weakening → resistant niche expansion

local constraints ← escape → amplification → therapy remodeling →

D Lévy-like spatial escape transitions



E Branch-mediated relapse amplification



spatial recolonization and therapy-resistant ecosystem emergence

● therapy-sensitive malignant state    ● resistant/relapse state    ● immune cell    ● stromal cell    — therapy / Lévy escape